

Oliver John (OJ) Watson

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EMPLOYMENT

Imperial College London **Oct 2023 – Present**

- Associate Professor in Infectious Disease Modelling and Health Emergencies; Eric and Wendy Schmidt AI in Science Fellow

London School of Hygiene and Tropical Medicine **Oct 2021 – Sep 2023**

- Schmidt Science Fellow

Imperial College London **Apr 2020 – Sep 2021**

- Postdoctoral Researcher in infectious disease modelling

Brown University **Oct 2019 – Mar 2020**

- Postdoctoral Researcher in computational genetics, bioinformatics

EDUCATION

Imperial College London **Oct 2015 – Sept 2019**

PhD Wellcome Trust Funded 1+3 PhD Student (October 2016 – October 2019. Viva Date 18 Nov 2019):

- Thesis Title: “Integrating genetic information into malaria transmission modelling”
- Supervisors: Prof Azra Ghani, Dr Lucy Okell and Dr Robert Verity

MRes Biomedical Research (2015 – 2016): Dissertation

- Dean’s Prize Awardee for top overall mark within MRes

University of Cambridge, Pembroke College **Oct 2011 – Jun 2015**

MSci Systems Biology (2014 – 2015): 1st Class (Top bioinformatics research project mark)

BA Natural Sciences (2011 – 2014): 1st Class (Foundation and College Scholarships for examination results)

SCIENTIFIC PROFILE

Evidence of Esteem

- 72 publications, 16 first/equal-contribution (*Science, Lancet Infectious Diseases, BMJ*), 2 book chapters
- 20 invited talks and 16 conference presentations (2 Invited, 5 Oral and 9 poster)
- Undergraduate, Masters and PhD Supervision experience (Brown, Oxford) and Masters Examination (Oxford, Imperial)
- Track record of securing fellowship funding, contributing to international interdisciplinary grants (3 NIH grants, US CDC, CEPI, WHO) and securing grants as principal investigator (Wellcome).

Committee and Consortium Memberships

- **WHO pfhrp23 and MFT Technical Panels (2023 - Present):** Supporting the update of the WHO response plan to pfhrp23 deletions, providing expertise and modelling to support malaria diagnostic policy. Additionally supporting WHO Technical panels on Multiple First Line Therapies (MFT).
- **ROpenSci (2018 - Present):** Community of software developers building open source and reproducible tools using the R programming language. I have contributed R packages and code reviews for data access and survey tools.
- **VacSafeWG (2022 - Present):** Schmidt Futures funded group for furthering vaccine safety and confidence-building in Africa. I contribute to policy briefs on vaccines, data sources and early warning systems for supranational agencies.

Academic Service

- **Journal Reviewer:** Nature, PNAS, Nature Communications, PLoS Computational Biology, Genome Biology and Evolution, PLoS Global Public Health, American Journal of Tropical Medicine & Hygiene, Evolutionary Applications, Malaria Journal, Conflict and Health, Royal Society Interface, Lancet Regional Health
- **Grant Reviewer:** Wellcome, NIH, Netherlands Organisation for Health Research and Development, Imperial College London (Internal Fellowship Rounds)

RESEARCH SUPPORT

Grants and Fellowships

As Principal Investigator

The UK Academy of Medical Sciences Springboard. Dr Oliver Watson (PI). Deep Learning Surrogates of Infectious Disease Models. £125,000.00.	2025 - 2027
MRC GIDA Pump Priming Funding. Dr Oliver Watson (PI). Hackathon - Bridging the Gap: Applied Infectious Disease Epidemiology Tools to Support Evidence-Informed Decision-Making. £31,151.49.	2024-2025
Community Jameel. Dr Oliver Watson (PI). Jameel Institute Real-time Intelligent Support for Emergencies (RISE): Vaccine and Mortality Modelling in the Gaza Conflict. £1,400,000.00.	2024 - 2026
Wellcome Trust. Dr Oliver Watson (PI). Leveraging forecasts to optimise decision making. £79,365	2024 - 2025
Imperial College Research Fellowship supported by an Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship. Dr Oliver Watson (PI). Novel methods and data sources for pandemic preparedness. £337,465	2023 - 2027
Schmidt Science Fellowship. Dr Oliver Watson (PI). Vaccine impact assessments and mortality estimation from COVID-19. \$200,000.	2021 - 2023
Wellcome Trust. Mr Oliver Watson (PhD Funding). Integration of parasite genetic information in malaria transmission modelling. £165,784.00	2015 - 2019

As Co-Investigator

MRC Mid Equipment Grant. Dr Rich FitzJohn (PI). Accelerating Infectious Disease Modelling Using General Purpose GPUs. £186,400.00. Named co-investigator.	2025 - 2026
Coalition for Epidemic Preparedness Innovations. Prof Azra Ghani (PI). Assessing and predicting the public health benefits and economic health impact of CEPI's investments for COVID-19. £40,000	2022
US CDC, U01GH002319. Prof Francesco Checchi (PI). Covid-19 and related public health threats in populations affected by crises: a multi-disciplinary, collaborative research programme. \$15,000,000 (Initial Award. Annual funder obligations reviewed through cooperative agreement). Named co-investigator, contribution in kind to ensure LMIC partner funding in response to funding reductions	2022 - 2027
NIH, 1R01AI177791-01A1. Dr Jonathan Parr (PI). Epidemiology and determinants of emerging artemisinin resistant malaria in Ethiopia. \$3,603,465. Named co-investigator. \$42,000 over 3 years.	2023 - 2028
NIH, 1R01AI156267-01A1. Dr Jeffrey Bailey (PI). Artemisinin Resistance in Africa: its emergence and evolution in Rwanda. \$3,989,037. Named co-investigator. \$60,000 over 3 years	2021 - 2026
NIH, 1R03AI173899-01A1. Dr Jessica Lin (PI). Does treating low density malaria infections reduce malaria transmission? \$100,000. Named co-investigator. \$15,000 over 2 years	2023 - 2025
World Health Organisation. Dr Alexandra Hogan (PI). Modelling the Impact of COVID-19 Vaccination Strategies. \$100,533. Drafted submission. \$52,325 allocated for my 100% FTE on grant as named researcher.	2021
Wellcome Trust. Dr Patrick Walker (PI). Modelling the global spread and determinants of COVID-19 transmission. £96,664. Drafted submission. £30,255 allocated for 100% FTE on grant as named researcher.	2020 - 2020

Invited Consulting

Clinton Health Access Initiative. Modelling the impact of alternative RDTs in sub-Saharan Africa	2024-2025
Clinton Health Access Initiative. Modelling the spread of antimalarial resistance in sub-Saharan Africa.	2023 - 2024
World Health Organization. Modelling the global risk and spread of pfhrp2/3 gene deletions.	2022 - 2023
Asian Development Bank. Evaluating COVID-19 vaccine impact and long term vaccination strategies in India.	2021 - 2022

* = Equal Author Contribution

1. Christen P, Dawa J, Ahmed O, Akinyi F, Ameyo B, Bekalo DB, Biwott C, Bosire AB, Charles G, Gathoni V, Gaythorpe KAM, Githinji G, Gowa S, Hamumy F, Kagiri H, Kamundia GM, Kamunya S, Kariuki P, Khalayi L, Kombe I, Kwamboka P, Lietar P, Likalamu N, Lugonzo B, Mariita V, Masinde A, Mbithi M, McCabe R, Monari FN, Motiri FO, Mutwiri MA, Mudamba AA, Mugo PW, Mutono N, Muriithi M, Musasia R, Mutunga M, Mwangi W, Nalwa J, Ngeno KI, Nyaberi JM, Nzoka PM, Ojal J, Okech T, Okunga E, Omondi E, Osoro J, Osoro P, Patel A, Sayianka S, Shigoli T, Silali C, Thendu MM, van Elsland SL, Wanjiku KA, Muchemi P, Whittles LK, Winskill P, Ombajo L, Thumbi SM, **Watson OJ**. Bridging the gap between public health academia and policy. *SocArXiv* (2025). , doi:10.31235/osf.io/tcjqs_v1
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3. Kraemer MUG, Tsui JL-H, Chang SY, Lytras S, Khurana MP, Vanderslott S, Bajaj S, Scheidwasser N, Curran-Sebastian JL, Semenova E, Zhang M, Unwin HJT, **Watson OJ**, Mills C, Dasgupta A, Ferretti L, Scarpino SV, Koua E, Morgan O, Tegally H, Paquet U, Moutsianas L, Fraser C, Ferguson NM, Topol EJ, Duchêne DA, Stadler T, Kingori P, Parker MJ, Dominici F, Shadbolt N, Suchard MA, Ratmann O, Flaxman S, Holmes EC, Gomez-Rodriguez M, Schölkopf B, Donnelly CA, Pybus OG, Cauchemez S, Bhatt S. Artificial intelligence for modelling infectious disease epidemics. *Nature*. 638, 623–635 (2025)
4. Wernsman Young N, Gashema P, Giesbrecht D, Munyaneza T, Maisha F, Mwebembezi F, Budodo R, Leonetti A, Crudale R, Iradukunda V, Jean Bosco N, Kirby RI, Boyce RM, Mandara CI, Kanyankole GK, Ntaro M, Okell LC, **Watson OJ**, Mulogo E, Ishengoma DS, Hangi S, Karema C, Mazarati J-B, Juliano JJ, Bailey JA. High frequency of artemisinin partial resistance mutations in the great lakes region revealed through rapid pooled deep sequencing. *J. Infect. Dis.* 231, 269–280 (2025)
5. Whittaker C, Barnsley G, Olivera Mesa D, Laydon DJ, Tan CW, Zhu F, Johnson R, Doohan P, Nedjati-Gilani G, Winskill P, Hogan AB, Deol A, Mukandavire C, Hauck K, Lye DCB, Wang L-F, **Watson OJ**, Ghani AC. Quantifying the impact of a broadly protective sarbecovirus vaccine in a future SARS-X pandemic. *medRxiv* (2024), doi:10.1101/2024.08.12.24311730
6. **Watson OJ**, Muchiri S, Ward A, Meier-Sherling C, Asua V, Katairo T, Brewer T, Cuomo-Dannenburg G, Winskill P, Bailey JA, Okell L, Scudu G, Woolsey AM. Risk of selection and timelines for the continued spread of artemisinin and partner drug resistance in Africa. *medRxiv* (2024), doi:10.1101/2024.08.28.24312699. <http://hdl.handle.net/10044/1/114684>
7. **Watson OJ**, Hogan AB. Impact of COVID-19 vaccination programmes in Europe: lives saved and lessons learned. *Lancet Respir. Med.* 12, 663–664 (2024)
8. Meier-Scherling CPG, **Watson OJ**, Asua V, Ghinai I, Katairo T, Garg S, Conrad MD, Rosenthal PJ, Okell LC, Bailey JA. Selection of *Plasmodium falciparum* kelch13 mutations in Uganda in comparison with southeast Asia: a modelling study. *Lancet Microbe*. 6, 101027 (2025)
9. **Watson OJ**, Tran TN-A, Zupko RJ, Symons T, Thomson R, Visser T, Rumisha S, Dzianach PA, Hathaway N, Kim I, Juliano JJ, Bailey JA, Slater H, Okell L, Gething P, Ghani A, Boni MF, Parr JB, Cunningham J. Global risk of selection and spread of *Plasmodium falciparum* histidine-rich protein 2 and 3 gene deletions. *Nature Medicine*. 2025;31:3372–3379. doi:10.1038/s41591-025-03974-3
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15. Sheppard RJ, **Watson OJ**, Pieciak R, Lungu J, Kwenda G, Moyo C, Chanda SL, Barnsley G, Brazeau NF, Gerard-Ursin ICG, Olivera Mesa D, Whittaker C, Gregson S, Okell LC, Ghani AC, MacLeod WB, Del Fava E, Melegaro A, Hines JZ, Mulenga LB, Walker PGT, Mwananyanda L, Gill CJ. Using mortuary and burial data to place COVID-19 in Lusaka, Zambia within a global context. *Nat. Commun.* 14, 3840 (2023)
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19. Andagalu B, **Watson OJ**, Onyango I, Opot B, Okoth R, Chemwor G, Sifuna P, Juma D, Cheruiyot A, Yeda R, Okudo C, Wafubwa J, Yalwala S, Abuom D, Ogutu B, Cowden J, Akala HM, Kamau E. Malaria Transmission Dynamics in a High-Transmission Setting of Western Kenya and the Inadequate Treatment Response to Artemether-Lumefantrine in an Asymptomatic Population. *Clin. Infect. Dis.* 76, 704–712 (2023)
20. Andagalu B, **Watson OJ**, Onyango I, Opot B, Okoth R, Chemwor G, Sifuna P, Juma D, Cheruiyot A, Yeda R, Okudo C, Wafubwa J, Yalwala S, Abuom D, Ogutu B, Cowden J, Akala HM, Kamau E. Reply to Blanken et al. *Clin. Infect. Dis.* 76, 557–559 (2023)
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22. **Watson OJ***, Gao B*, Nguyen TD*, Tran TN-A, Penny MA, Smith DL, Okell L, Aguas R, Boni MF. Pre-existing partner-drug resistance to artemisinin combination therapies facilitates the emergence and spread of artemisinin resistance: a consensus modelling study. *Lancet Microbe.* 3, e701–e710 (2022)
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28. Favas C, Jarrett P, Ratnayake R, **Watson OJ**, Checchi F. Country differences in transmissibility, age distribution and case-fatality of SARS-CoV-2: a global ecological analysis. *Int. J. Infect. Dis.* 114, 210–218 (2022)
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66. **Watson OJ***, Sumner KM*, Janko M, Goel V, Winskill P, Slater HC, Ghani A, Meshnick SR, Parr JB. False-negative malaria rapid diagnostic test results and their impact on community-based malaria surveys in sub-Saharan Africa. *BMJ Glob Health.* 4, e001582 (2019)
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68. **Watson OJ**, Verity R, Ghani AC, Garske T, Cunningham J, Tshetu A, Mwandagaliwa MK, Meshnick SR, Parr JB, Slater HC. Impact of seasonal variations in *Plasmodium falciparum* malaria transmission on the surveillance of *pfrp2* gene deletions. *Elife.* 8, e40339 (2019)
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72. **Watson OJ**, Slater HC, Verity R, Parr JB, Mwandagaliwa MK, Tshetu A, Meshnick SR, Ghani AC. Modelling the drivers of the spread of *Plasmodium falciparum* *hrp2* gene deletions in sub-Saharan Africa. *Elife.* 6, e25008 (2017)

Book Chapters

Routledge I, **Watson OJ**, Griffin JT, Ghani AC. "Predictive Malaria Epidemiology, Models of Malaria Transmission and Elimination" in *Encyclopedia of Malaria*, P. G. Kremsner, S. Krishna, Eds. (Springer New York, New York, NY, 2018). pp. 1–7

Watson OJ, Routledge I, Griffin JT, Ghani AC. "Predictive Malaria Epidemiology, Models of Malaria Control Interventions and Elimination" in Encyclopedia of Malaria, P. G. Kremsner, S. Krishna, Eds. (Springer New York, New York, NY, 2018). pp. 1–7

REPORTS

What models are available for predicting infectious outbreaks in Africa?. Chapter Contribution for Signature Initiative to Mitigate Biological Threats in Africa. A partnership initiative between the Africa CDC and the G7 led Global Partnership

Against the Spread of Weapons and Materials of Mass Destruction. Under Africa CDC Approval

Update of the "Response plan to pfhrp2/3 gene deletions". Contributing author to WHO Response Plan. Forthcoming

Imperial College COVID-19 Reports. Author of 19 Imperial College COVID-19 Reports including (Co)-First/Last Author of 6 Reports, all of which have led to peer-reviewed scientific publications. 2020-2021

UK COVID-19 Public Inquiry. First Witness Statement of Professor Neil Ferguson. Contributed writing to three sections as part of UK COVID-19 Public Inquiry. July 2023

A preliminary assessment of COVAX's impact in lower-income countries. Contributed analysis to GAVI white paper about the impact of the COVAX Facility on lives saved due to COVID-19 vaccination in LMICs. May 2023

From insight to action: Examining mortality in Somalia. Contributing author to WHO and Unicef report on Excess Mortality in Somalia due to drought. March 2023

Mortality patterns in Somalia: Retrospective estimates and scenario-based forecasting. Lead author of modelling analysis underpinning WHO and Unicef Policy Report on Mortality in Somalia. March 2023

Evaluating COVID-19 vaccine impact and long term vaccination strategies in India. Lead Author of Asian Development Bank Internal Report. March 2022

COVID-19: Preparing for the future. Looking ahead to winter 2021/22 and beyond. Contributed analysis to UK Academy of Medical Sciences Rapid Review. June 2021

An assessment of COVID-19 mortality in humanitarian contexts. Contributing Author to UNHCR COVID-19 Research. March 2021

SEMINARS AND INVITED PRESENTATIONS

Jameel Institute Real-time Intelligent Support for Emergencies (JI-RISE). Oral Presentation. Jameel Institute Annual Symposium. 20 November 2024

Mathematical modelling - a tool in evaluating drug policies and antimalarial resistance containment . Oral Presentation. WHO Technical consultation on multiple first-line therapies as part of the strategy to respond to antimalarial drug resistance. 15 May 2024

Timelines for the continued spread of artemisinin and partner drug resistance in Africa . Oral Presentation. WHO Steering Committee Meeting on Antimalarial Resistance. 9 April 2024

Experience of being a Schmidt Science Fellow. Panel Member. Rhodes Trust Trustee Meeting. Q&A with fundees panel event. 8 March 2024

Equitably equipping the response to conflicts and pandemics. Oral Presentation. Schmidt Science Fellows Alumni Conference, Toronto. 18 July 2024

How to use ChatGPT for writing modelling papers. Oral Presentation. Imperial College London. Seminar Training Series. 20 June 2024

Modeling to understand the past and predict the future. Oral Presentation. INES Runhegeri, Rwanda. 29 April 2024

International Advances in COVID-19. Framework for Scientific Evolution and Evaluation. Oral Presentation. UCSF Fresno Town Hall Lecture Series. 7 December 2023

Surrogate modelling of agent based model features: an application for evaluating the risk of malaria parasites evolving to evade diagnosis. Seminar. University of Bergen. Workshop on teaching and research in statistics and data science. 28 November 2023

Approaches to Navigating Science Beyond 2050. Chair of Panel Session. Schmidt Science Fellows 5 year Alumni Conference. 18 November 2023

Update on the state of COVID-19 underreporting in Africa. Afternoon. The impact of pfhrp2/3 deletions on future testing of malaria in Africa. Seminar. Johns Hopkins University. 21 July 2023

Global risk of selection and spread of Plasmodium falciparum histidine-rich protein 2 and 3 gene deletions: Implications for future RDT procurement and WHO malaria control policy guidance. Oral Presentation. President's Malaria Initiative RDT Task Force. 10 July 2023

Mortality patterns in Somalia: retrospective estimates and scenario-based forecasts. Oral Presentation. Somalia Ministry of Health; WHO EMRO; Horn of Africa Food Security and Nutrition Working Group. 30 March 2023

Risk based approach to transitioning to alternative RDTs. Oral Presentation. WHO Meeting to update response plan to pfhrp2/3 gene deletions. 26 January 2023

Pros and Cons of Global Roll-Out. Oral Presentation. International Workshop on COVID-19 Vaccines 2022 - December Edition organised by Virology Education. 15 December 2022

Integrating novel data sources for COVID-19 outbreak modelling. Seminar. Afya Research Consortium. Nairobi. 20 July 2022

COVID-19 Vaccines: Unfinished Business. Chair of Panel Session. International Symposium on Vaccines and Global Health. 1 April 2022

Lessons in persistence for fellowship rejections and practising equanimity in academia. Oral Presentation. Johns Hopkins University Lunchtime Seminar. 29 March 2022

Utilising vaccine safety and confidence data in support of modelling global vaccine roll out. Oral Presentation. Inaugural meeting of Columbia-Wits Vaccine Safety and Confidence-Building Working Group. 29 April 2021

Imperial SPI-M & covidsim.org Modelling Overview. Seminar. UK Cabinet Office. 16 March 2021

Using Earth Observation to understand the COVID-19 pandemic. Oral Presentation. Imperial Earth Observation Network. 25 February 2021

Impact of COVID-19 vaccination in India: a subnational mathematical modelling study. Oral Presentation. Asian Development Bank. 20 January 2021

Modelling COVID-19 epidemic in Khartoum. Oral Presentation. Sudan Federal Ministry of Health. 9 December 2020

COVID-19 Mortality Underreporting: Investigation into Damascus, Syria. Oral Presentation. Save The Children COVID-19. 16 November 2020

COVID-19 Mortality Underreporting: Investigations in Damascus, Khartoum and Aden. Oral Presentation. WHO COVID-19 Modeling Group. 23 October 2020

Modelling to support COVID-19 decision making. Oral Presentation. UK DfID Country Offices. 25 June 2020

Short-term forecasts of healthcare demand for countries in the early stage of COVID-19 epidemics. Oral Presentation. WHO COVID-19 Modeling Group. 15 May 2020

Infectious disease modelling 101- An introduction to COVID-19 models. Seminar. PATH. 14 April 2020

Integration of malaria parasite genetics into malaria transmission modelling. Seminar. Warren Alpert Medical School of Brown University. 22 October 2019

Modelling the epidemiological implications and explanations of discordances between microscopy and rapid diagnostic tests. Seminar. University of North Carolina at Chapel Hill. 1 November 2017

CONFERENCE AND MEETING PRESENTATIONS

Oral Presentation. Hamlet A, ... **Watson OJ**. Jameel Institute Crisis Vaccination Planner: Prototype Modelling Tool to Prioritise Vaccination in Humanitarian Emergencies. MSF Scientific Day, London. 2025

Symposium Chair and Oral Presentation. **Watson OJ** et al. Artemisinin Resistance Response in Africa: Integrating Molecular Surveillance and Mathematical Modeling to Mitigate Emerging Risk and Impact on Malaria Burden Reduction. ASTMH. 2024

Oral Presentation. **Watson OJ** et al. Risk of selection and timelines for the continued spread of artemisinin and partner drug resistance in Africa. Multilateral Initiative On Malaria (MIM Society) 8th Pan-African Malaria Conference. 2024

Symposium Presentation. **Watson OJ** et al. Global risk of selection and spread of Plasmodium falciparum histidine-rich protein 2 and 3 gene deletions: Implications for future RDT procurement and malaria control policy guidance. Multilateral Initiative On Malaria (MIM Society) 8th Pan-African Malaria Conference. 2024

Poster Presentation. Barnsley G, ..., **Watson OJ**. Modelling the potential impact of 100 days mission and broader investments on the COVID-19 pandemic. 17th Vaccine Congress. 2023

Poster Presentation. Meier-Scherling C, **Watson OJ** et al. Comparison of strength of selection for Plasmodium falciparum artemisinin resistance-associated mutations between southeast Asia and Uganda. ASTMH. 2023

Poster Presentation. McCabe R, ..., **Watson OJ**. Alternative epidemic indicators for COVID-19 in three settings with incomplete death registration systems. ASTMH. 2023

Oral Presentation. Barnsley G, ..., **Watson OJ**. Modelling the potential impact of 100 Days Mission and broader investments on the COVID-19 pandemic. Epidemics 9. 2023

Invited Meeting Participant. Rhodes Policy Summit: Creating a Positive Legacy from the Pandemic. London. 2023

Invited Chair of Panel Session. Karlinsky A, **Watson OJ** et al. The old-new science of excess mortality. Royal Statistical Science Conference. 2023

Oral Presentation. **Watson OJ** et al. Vaccine Impact Estimation: Pros and Cons of the Global COVID-19 Vaccination Campaign and evaluation of improved R&D in the Republic of Korea. UK-Korea bilateral international meeting organised by The Royal Society. 2023

Poster Presentation. Barnsley G, ..., **Watson OJ**. Global impact of the first year of COVID-19 vaccination: a mathematical modelling study. 16th Vaccine Congress. 2022

Poster Presentation. Paschalidis A, **Watson OJ** et al. Complexity of Infection Estimation with Allele Frequencies. ASTMH. 2021

Oral Presentation. Okell L, **Watson OJ** et al. Multiple first line therapies versus reducing overprescription of antimalarials to slow antimalarial resistance. ASTMH. 2018

Poster Presentation. **Watson OJ** et al. The impact of seasonal variation in the detection of clinically relevant plasmodium falciparum hrp2 gene deletions: a modelling study. ASTMH. 2018

Poster Presentation. **Watson OJ** et al. Evaluating the performance of malaria genomics for inferring changes in transmission intensity using transmission modelling. Genomic Epidemiology of Malaria. 2018

Poster Presentation. Cremin I, **Watson OJ** et al. An infectious way to teach students about outbreaks. Epidemics 6. 2017

Oral Presentation. **Watson OJ** et al. rdhs: an R package to interact with the demographic and health surveys (DHS) program data sets. ASTMH. 2017

Oral Presentation. **Watson OJ** et al. Characterizing the potential bias within genomic tools for inferring changes in plasmodium falciparum transmission intensities. ASTMH. 2017

Poster Presentation. **Watson OJ** et al. Modelling the epidemiological implications and explanations of discordances between microscopy and rapid diagnostic tests. Future of Malaria Research Symposium. 2017

Oral Presentation. **Watson OJ** et al. Drivers of the spread of “diagnostic resistant” P. falciparum malaria: a model-based evaluation of the spread of pfhrp2 gene deletions in Africa. ASTMH. 2016

Invited Oral Presentation. **Watson OJ** et al. Modelling the drivers of Plasmodium falciparum hrp2 deletions. WHO organised panel meeting on “Plasmodium falciparum hrp2/3 gene deletions: update, implications and response” at ASTMH. 2016

TEACHING EXPERIENCE

PhD Supervision:

Cecile Meier-Scherling. Brown University. Comparison of strength of selection for *P. falciparum* artemisinin resistance-associated mutations between southeast Asia and Uganda. 2022 -

Ruth McCabe. University of Oxford. Epidemiological analysis and modelling of COVID-19 and other high consequence infectious disease transmission risks. Expected 2023

Masters Supervision:

MSc Epidemiology. Hallie Benjamin. LSHTM. Return on Investment of COVID-19 Vaccines. 2024

MRes Biomedical Research, Epidemiology, Evaluation and Control of Infectious. Chaokun Hong. Modelling the impact of drug resistance on the selection of resistance to front line antimalarial treatment. 2024

MSc Epidemiology. Marco Chan Yan Chit. Modelling the impact of drug resistance on the selection of resistance to front line antimalarial treatment. 2024

Undergraduate Supervision:

Rebecca Kirby. Brown University. Examining the Early Distribution of the Artemisinin-Resistant *Plasmodium falciparum* kelch13 R561H Mutation in Areas of Higher Transmission in Rwanda. Published in Open Forum Infectious Diseases, Volume 10, Issue 4, April 2023, ofad149. 2020 - 2021

Aris Paschialidis. Brown University. Direct estimation of COI from whole genome sequence data. Published in PLoS Computational Biology. 2023;19(6):e1010247. 2020 - 2022

Teaching:

Module lead on 16-week Health Protection Module. Imperial College London GMPH. 2024-

Weekly distance learning for 5 Somali researchers at SIMAD university for mortality estimation in R. 2023

Demonstrator on 2-day practical on developing a model to answer a public health control policy question. Imperial College London. MSc Epidemiology. 2016 - 2021

Introduction to R and statistical modelling. 3 day introductory course for computational biology rotation students. Brown University. 2020

Organised teaching material and demonstration assistance on Imperial College London 2-week short course on infectious disease modelling. 2016 - 2018

Prepared computer-based practical exercise on outbreak reconstruction. Imperial College London. MSc Epidemiology. The practical exercise and its administration were subsequently published as a pedagogical exercise ("An infectious way to teach students about outbreaks". *Epidemics*. 23, 42–48 (2018)) 2016

Demonstrator on the use of multi locus sequence typing databases in epidemiology. Imperial College London. MSc Epidemiology 2016

For my teaching demonstrations, I have been nominated for a Student Academic Choice Awards (SACA) Teaching Award (2019).

I am passionate about improving my supervision approach and mentorship and have undertaken both external training on career development offered by Stanford University and provide evaluation forms from all students and researchers I supervise and manage (which have all been scored as excellent (6 out of 6), N = 5). This was an initiative I chose to start, both to understand how I can improve as I develop seniority and more frequently supervise other researchers but also in response to expanding my research supervision and support across time zones and country borders.

SOFTWARE DEVELOPMENT

I am an author of dozens of reproducible research compendia (committed to open, reproducible science) and the developer and maintainer of multiple R packages, with the top 8 most used frequently cited in studies by researchers around the world.

rdhs. API Client and Data Munging for the Demographic and Health Survey Data. Cited in 27 publications. <https://github.com/ropensci/rdhs>

squire. SEIR transmission model of COVID-19 and model fitting tools. Cited in 14 publications. <https://github.com/mrc-ide/squire>

nimue. Extension to the squire package that allows vaccination to be included. Cited in 6 publications. <https://github.com/mrc-ide/nimue>

McCOILR. R implementation of The REAL McCOIL method for estimating complexity of infection in malaria infections. Cited in 3 publications. <https://github.com/OJWatson/McCOILR>

hrp2malaria. Individual based model of Imperial College malaria model for modelling hrp23 deletions. Cited in 3 publications. <https://github.com/OJWatson/hrp2malaria>

safir. Individual based rewrite of the squire and nimue COVID-19 models. Cited in 2 publications. <https://github.com/mrc-ide/safir>

outbreakteachR. Accompanying R package for assisting in the analysis and execution of a simulated outbreak amongst students. Cited in 1 publication. <https://github.com/mrc-ide/outbreakteachR>

magenta. Spatial, individual based model of Imperial College malaria model for modelling parasite genetics. Cited in 1 publication. <https://github.com/OJWatson/magenta>

PROFESSIONAL MEMBERSHIP AND SERVICE

Reviewer. Nature, PNAS, Nature Communications, PLoS Computational Biology, Genome Biology and Evolution, PLoS Global Public Health, American Journal of Tropical Medicine & Hygiene, Evolutionary Applications, Malaria Journal, Conflict and Health, Royal Society Interface, Lancet Regional Health

Grant Reviewer. Wellcome Trust, NIH, Netherlands Organisation for Health Research and Development, Imperial College London (Internal Fellowship Rounds)

Shadow Interview Panels. Imperial College London (2023)

Interview Panels. Imperial College London (2021, '22, '23), LSHTM (2022, '23)

Examiner. PhD Advisory Committee (Temple, UNC at Chapel Hill), Imperial PhD (Early Stage Assessment), Imperial MRes (2024), Imperial GMPH (2024), Imperial MSc (2024, 2021), Imperial MPH (2024, 2021), Oxford MSc (2021, '23)

Member. Afya Consortium: US CDC funded consortium of academic and civil society organisations, working equitably to strengthen research capacity, improve data equity and conduct research in humanitarian settings.

Member. WHO pfhrp23 technical panel: Supporting the update of the WHO response plan to pfhrp23 deletions, providing expertise and modelling to support malaria diagnostic policy.

Member. WHO MFT technical panel: Supporting the update of the WHO malaria policy guidance on multiple first line therapies, providing expertise and modelling to support malaria drug policy.

Member. ROpenSci: Community of software developers building open source and reproducible tools using the R programming language. I have contributed R packages and code reviews for data access and survey tools.

Member. VacSafeWG: Schmidt Futures funded group for furthering vaccine safety and confidence-building in Africa. I contribute to policy briefs on vaccines, data sources and early warning systems for supranational agencies.

PUBLIC ENGAGEMENT AND OUTREACH

Various media appearances including TV (Sky, BBC, Euronews, TV Globo), Radio (BBC, RTS, Sky) and News (FT, Times, Guardian, Washington Post, NY Times, Economist - cover story) related to my COVID-19 work. Additionally, I provide to a number of fact-checking agencies (AFP Fact Check, German Press Agency dpa).

Science in Context Imperial College London Videos to discuss findings from our research.

Facebook Live Community Outreach with Sudanese Health Influencers. Conducted a series of live interviews and AMA with an Arabic interpreter to explain the impacts of our research in Khartoum (2020).

RISE Residential Summit. Interactive classroom simulation and pandemic tabletop exercise showcasing the role of infectious disease modelling in policy making. Delivered to the 100 winners of the RISE Global Scholarship program (2023).

Central London Data Science. Organiser of a meetup group teaching fundamentals of data science to over 3000 members across industry, academia and members of the public. (Founded in 2016).

Brilliant Club Course Author. Co-authored an 8-week Key Stage 2 UK (ages 7-11) curriculum entitled "Disease Detectives" (2018).